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Part A

1. Write short notes on RTD.

[3 marks] (December 2021)

Resistance temperature detectors, or resistance thermometers, employ a sensitive element of extremely pure platinum, copper or nickel wire that provides a definite resistance value at each temperature within its range.

The relationship between temperature and resistance of conductors in the temperature range near 0°C can be calculated from the equation

$$R_t = R_{ref} (1 + \alpha \Delta t)$$

where R_t = resistance of the conductor at temperature t (°C)

R_{ref} = resistance at the reference temperature, usually 0°C

α = temperature coefficient of resistance

Δt = difference between operating and reference temperature

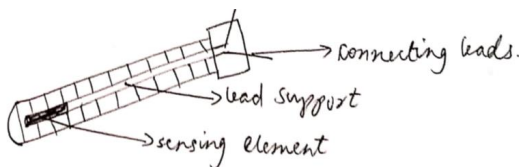
This change in resistance (ΔR) can be measured with a Wheatstone bridge, which may be calibrated to indicate the temperature that caused the resistance change rather than the resistance change itself

Resistance thermometers are generally of the probe type for immersion in the medium whose temperature is to be measured or controlled.

A typical sensing element for a probe-type thermometer is constructed by coating a small platinum or silver tube with ceramic material, winding the resistance wire over the coated tube, and coating the finished winding again with ceramic.

This small assembly is then fired at high temperature to assure annealing of the winding and then it is placed at the tip of the probe.

The probe is protected by a sheath to produce the complete sensing elements as shown in Figure

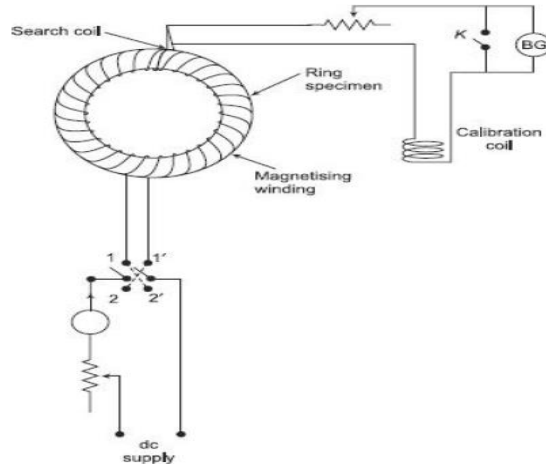


2 Describe anyone method of permeability measurement. [3 marks] (December 2021)

A ring specimen with known dimensions is used for the test.

A search coil is wound over the specimen. This search is known as a “B-coil”. This search coil is then connected to a ballistic galvanometer.

The arrangement of the test circuit is shown in Figure



After de-magnetizing the test is started by setting the magnetizing current to its lowest test level.

At that time, the galvanometer switch is closed, the iron specimen is brought into a reproducible cyclic magnetic state by throwing the reversing switch S backward and forward about twenty times.

The K switch is now opened and the value of flux corresponding of this value of H is measured by reversing the switch S and recording the throw of galvanometer.

The value of flux density corresponding to this H can be calculated by dividing the flux by the cross-section area of the specimen.

Then the ratio B/H gives the permeability.

3. What are the principal requirements in magnetic measurements?

[3 marks] (December 2020)

The operating characteristics of electrical machines, apparatus and instruments are greatly influenced by the properties of Ferro-magnetic materials used for their construction. Therefore, magnetic measurements and a thorough knowledge of characteristics of magnetic materials are of utmost importance in designing and manufacturing electrical equipment.

The principal requirements in magnetic measurements are

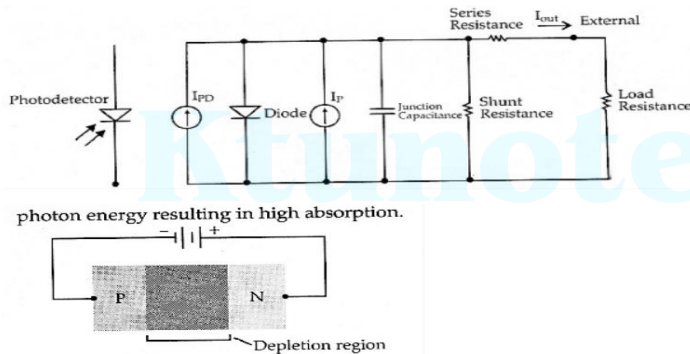
- (i) The measurement of magnetic field strength in air.
- (ii) The determination of B-H curve and hysteresis loop for soft Ferro-magnetic materials.
- (iii) The determination of eddy current and hysteresis losses of soft Ferro-magnetic materials subjected to alternating magnetic fields.
- (iv) The testing of permanent magnets.

Magnetic measurements have some inherent inaccuracies due to which the measured values depart considerably from the true values. The inaccuracies are due to the following reasons

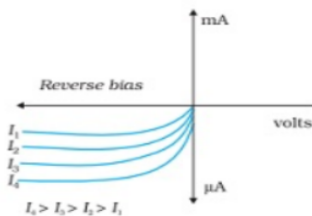
- (i) The conditions in the magnetic specimen under test are different from those assumed in calculations;
- (ii) The magnetic materials are not homogeneous
- (iv) There is no uniformity between different batches of test specimens even if such batches are of the same composition.

4. With a neat circuit explain the characteristics of photodiode. [3 marks] (December 2020)

A photodiode is a semiconductor device that converts light energy to electrical energy (voltage or current). The other name of the photodiode is photo-detector, photo-sensor, or light detector. The positive terminal of the supply is connected to the N terminal, and the negative side of the battery is connected to the P terminal. It is a highly sensitive diode to light so when light or photons falls on it. It converts the light to electrical energy.



$I - V$ characteristics: The curve shows the variation in current with the reverse bias voltage for the light of intensities I_1 , I_2 and I_3 falls on it.

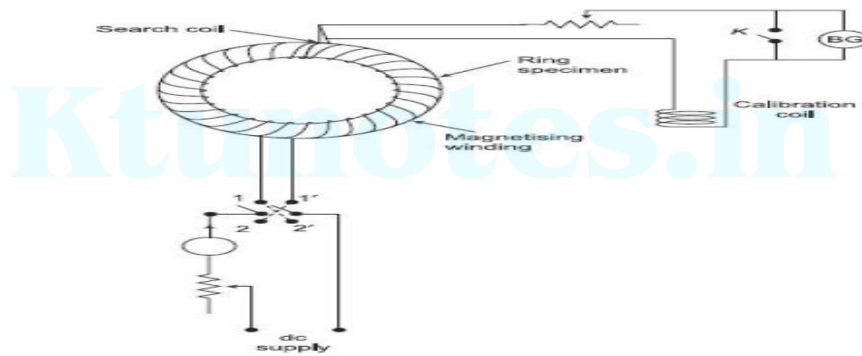


PART B

1. Explain the method of flux measurement using a ballistic galvanometer. [6 marks] (December 2021)

Measurement of flux density inside a specimen can be done by winding a search coil over the specimen.

- This search is known as a “B-coil”. This search coil is then connected to a ballistic galvanometer or to a fluxmeter.
- Scheme of measuring the flux density in a ring specimen is shown in Figure .
- Let, the current I is flowing through magnetizing winding wound around the specimen.
- A search coil with suitable number of turns is wound around the specimen and connected through a resistance and calibrating coil, to a ballistic galvanometer.
- The current through the magnetizing coil is reversed and therefore the flux linkages of the search coil change inducing an emf in it.
- This emf sends a current through the ballistic galvanometer causing it to deflect.



ϕ = flux linking the search coil

R = resistance of the ballistic galvanometer circuit

N = number of turns in the search coil

and t = time taken to reverse the flux

Average emf induced in the search coil

$$e = N \frac{d\phi}{dt} = 2 \frac{N\phi}{t}$$

So, average current through the ballistic galvanometer is,

$$i = 2N \frac{\phi}{Rt}$$

Charge passing through it,

$$Q = i \cdot t = \frac{2N\phi}{R}$$

Let θ_1 be the deflection of the galvanometer. Charge indicated by the galvanometer = $K_2 \theta_1$

So,
$$\frac{2N\phi}{R} = K_2 \theta_1$$

or flux
$$\phi = \frac{RK_2 \theta_1}{2N}$$

2. Describe the construction and working of photoconductive and photovoltaic cells. [8 marks]

(December 2021)

Photo conductive cell

The photoconductive cell converts the light energy into an electric current.

It uses the semiconductor material like cadmium selenide, Ge, Se, as a photo sensing element. When the beam of light falls on the semiconductor material, their conductivity increases and the material works like a closed switch.

The current starts flowing into the material and deflects the pointer of the meter. It works on photoconductive effect.

In this effect, the electrical resistance of the material varies with the amount of incident light, as shown in Fig. (b).
A typical construction is as shown in Fig. (a). The photo conductive material, typically Cadmium sulphide, Cadmium selenide or Cadmium sulpho-selenide, is deposited in a zig zag pattern (to obtain a desired resistance value and power rating) separating two metal coated areas acting as electrodes, all on an insulating base such as ceramic. The assembly is enclosed in a metal case with a glass window over the photo conductive material.

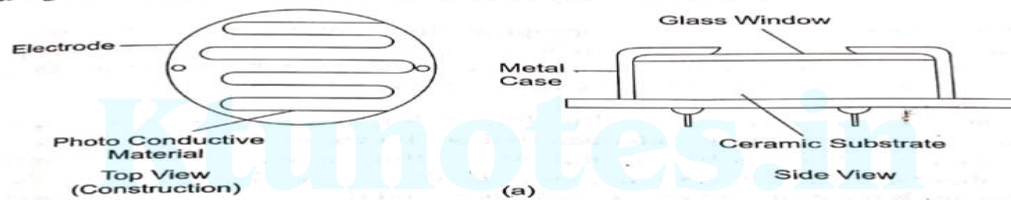
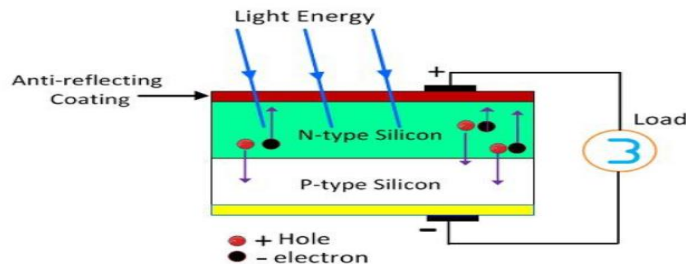


Photo-voltaic cell

The photovoltaic cell is the type of active transducer. The current starts flowing into the photovoltaic cell when the load is connected to it. The silicon and selenium are used as a semiconductor material. When the semiconductor material absorbs heat, the free electrons of the material starts moving. This phenomenon is known as the photovoltaic effect.



Photovoltaic Cell

Circuit Globe

The movements of electrons develop the current in the cell, and the current is known as the photoelectric current.

3. Explain in detail the measurement of iron losses in a magnetic material using Lloyd-Fisher square wattmeter method.. [8 marks] (December 2021)

2. Lloyd-Fisher Square. This is the most commonly used magnetic square and therefore it is described in greater details. The strips used are usually 0.25 m long and 50 to 60 mm wide. These strips are built up into four stacks. Each stack is made up of two types of strips one cut in the direction of rolling and the other cut perpendicular to the direction of rolling.

The stacks of strips are placed inside four similar magnetising coils of large cross-sectional area. These four coils are connected in series to form the primary winding.

Each magnetising coils has two similar single layer coils underneath it. They are called secondary coils. Thus in a magnetic square there are eight secondary coils. These secondary coils are connected in series in groups of four, one from each core, to form two separate secondary windings.

Test set up. The test specimen is weighed before assembly and its effective cross-section is determined. The primary winding is connected to a sinusoidal voltage supply. Fig. shows the connection diagram for finding the total iron loss by wattmeter method.

The primary winding contains the current coil of the wattmeter. The pressure coil of the wattmeter is supplied from one of the secondary windings. The wattmeter is designed for low power factor operation as the power factor is usually about 0.2. The second secondary winding supplies an electrostatic voltmeter (or an electrodynamic voltmeter of very high impedance). The frequency of supply is adjusted to the correct value.

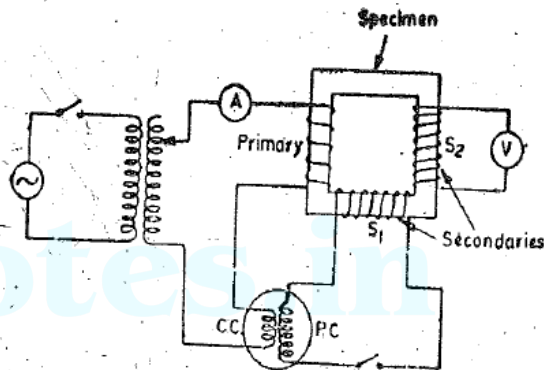


Fig. ... Wattmeter method for determining iron loss.

The voltage applied to the primary winding is adjusted, preferably with the help of a variable ratio transformer, till the magnetising current is adjusted to give the required value of B_m . The wattmeter and voltmeter readings are observed.

Theory. The electrostatic voltmeter connected across secondary winding S_2 measures the rms value of induced voltage. This value of induced voltage is

$$E = 4 K_f \Phi_m f N_2 = 4 K_f B_m A_s f N_2$$

where

K_f = form factor (1.11 for sinusoidal waves),

Φ_m = maximum value of flux ; Wb,

B_m = maximum value of flux density ; Wb/m²,

A_s = cross-sectional area of specimen ; m²,

f = frequency ; Hz,

and

N_2 = number of turns in the secondary winding S_2 .

$$\therefore \text{Maximum value of flux density } B_m = \frac{E}{4 K_f A_s f N_2}$$

This value is thus the apparent value B_m' of the flux density.

The wattmeter reading includes both the iron loss in the specimen and the copper loss in the secondary winding circuit. The copper loss in the secondary winding can be calculated and subtracted from the wattmeter readings.

Let P_i = total iron loss, P = wattmeter reading,

V = voltage applied to wattmeter pressure coil,

E = voltmeter reading = voltage induced in S_2 ,

r_p = resistance of wattmeter pressure coil, r_c = resistance of coil S_1 ,

and I_p = current in the pressure coil circuit.

Now, voltage induced in S_1 is equal to the voltage induced in S_2 since both of them have equal number of turns and they link with the same flux.

$\therefore$ Voltage induced in $S_1 = E$

The leakage reactances of winding S_1 and that of pressure coil are very small as compared with resistance of the pressure coil circuit. If we neglect the leakage reactances of winding S_1 and pressure coil, we have

$$E = I_p (r_p + r_c)$$

$\therefore$ Total iron loss in specimen + total copper loss in the secondary winding circuit = PE/V

Total copper loss in the secondary winding circuit = $E^2/r_p + r_c$

$$\therefore \text{Total iron loss in the specimen } P_i = P \frac{E}{V} - \frac{E^2}{r_p + r_c} = P \left(1 + \frac{r_c}{r_p} \right) - \frac{E^2}{r_p + r_c}$$

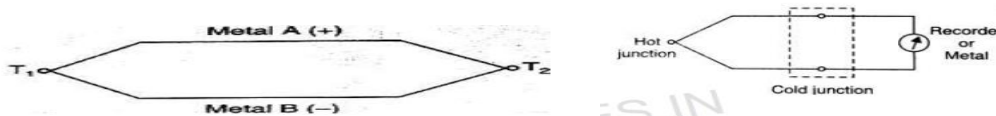
Specific iron loss i.e., iron loss per kg can be calculated by dividing the total iron loss by the weight of the specimen.

The eddy current and hysteresis components of the iron loss can be determined from results of the loss measurements at different frequencies.

4. Explain the temperature measurement using thermocouples. [6 marks] (December 2021)

Consists of a pair of dissimilar conductors welded or fused together at one end to form the hot or measuring junction with other end available for connection to the cold or reference junction

Converts thermal energy into electrical energy



When the hot junction temperature is greater as compared to the cold junction an emf is generated due to the temperature gradient.

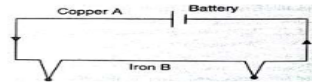
4.9.3.1 Thermoelectric Effects

1. Seebeck Effect.

- Two wires of different metals A and B are joined together to form two junctions and if the two junctions are at different temperatures, an electric current will flow around the circuit

Peltier Effect

- This is the reverse phenomenon of Seebeck effect.
- Suppose, the two junctions of a thermocouple are initially at the same temperature and electric current is passed through the circuit by using a battery. It is observed that heat is evolved at one end of the junction, which becomes hot, and heat is absorbed at the other end, which becomes cold. If the direction of current is reversed, the colder junction becomes hot and vice versa.



5. Explain with figure how BH Curve and Hysteresis loop can be determined using step by step method. [8 marks] (December 2020)

Step-by-step Method

For this test, the circuit diagram is shown in Figure below.

In this test the magnetization winding of the ring specimen is supplied from a potential divider having few tapings.

The tapings are arranged so that the magnetization force H may be increased in suitable number of steps, up to desired maximum value.

Before testing the specimen, it must be demagnetized.

Keeping the switch S_2 on the tap 1 position, the switch S_1 is closed.

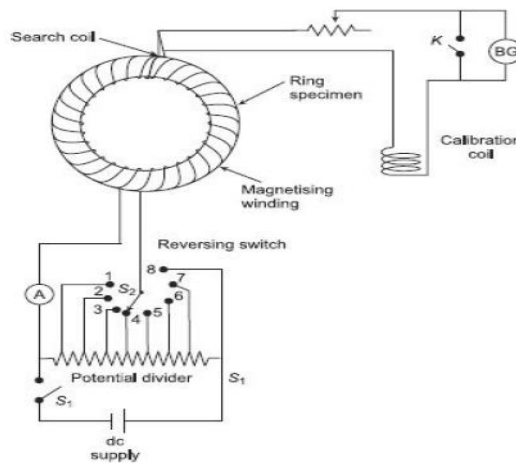


Figure *Circuit to measure magnetization curve*

Current flow in the magnetizing winding sets up a magnetizing force, H_1 which in turn increases the flux density in the specimen, from zero to some value B_1 , and corresponding throw of galvanometer is observed.

EET 203 Measurements & Instrumentation
Module IV: University Questions and Answers

Prepared by Dr. Renuka T K , EEE, MESCE

The B_1 value can be calculated from the throw of the galvanometer. H_1 value may be calculated from the reading of the ammeter, connected to the magnetizing winding circuit.

The magnetizing force is then increased to some value H_2 by switching S_2 suddenly to the tapping 2 and the corresponding increase in flux density ΔB is determined from the throw of the galvanometer.

Then the flux density B_2 corresponding to magnetizing force H_2 is calculated as $B_2 = B_1 + \Delta B$.

The process is repeated for other values of H up to the maximum point and the complete B - H curve thus

drawn as shown in Figure below.

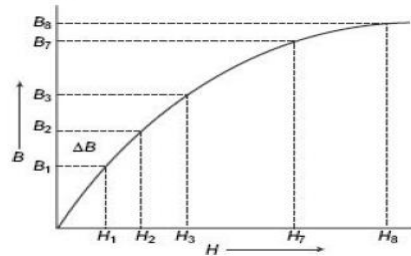


Figure 1 Magnetising curve

The determination of the hysteresis loop by this method is carried out simply by continuing the procedure described in the previous section where the B - H curve was obtained.

After reaching the point of maximum H with S_2 on the tapping 8, the magnetizing current is then reduced, in steps to zero by moving S_2 down through tapping points 8,7,6...2,1.

After the reduction of the magnetizing force to zero, negative values of H are obtained by reversing the reversing supply to the potential divider and then moving the switch S_2 in steps as before.

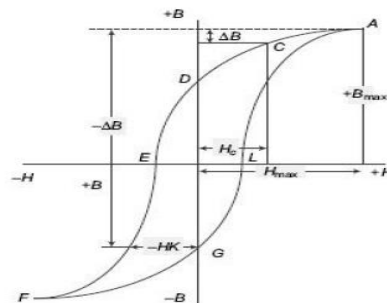


Figure 2 Hysteresis loop

6. List the salient features of thermistors.

[6 marks] (December 2020)

Thermistors : Thermally sensitive Resistors : NTC thermistor definition

An NTC thermistor is a thermally sensitive resistor whose resistance exhibits a large, precise and predictable decrease as the core temperature of the resistor increases over the operating temperature range.

Thermistors (construction of thermal resistor) are semiconductors which behave as resistors with a high negative temperature coefficient of resistance.

Manganese, nickel or cobalt oxides are milled, mixed in proper proportion with binders, passes into desired shapes and then sintered to form thermistors in the form of beads, rods or discs. Sometimes, a glass envelope is provided to protect a thermistor form contaminations.

The resistance R_T of a thermistor at temperature T (Kelvin) can be written as

$$R_T = R_0 \exp \left[\beta \left(\frac{1}{T} - \frac{1}{T_0} \right) \right]$$

where R_T and R_0 are the resistances in ohms of the thermistor at absolute temperatures T

and T_0 . β is a thermistor constant ranging from 3500 K to 5000 K. The reference temperature T_0 is usually taken as 298 K or 25°C.

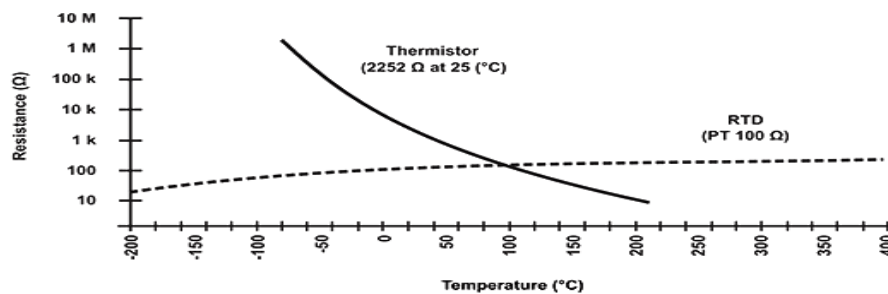
Characteristics of NTC thermistors

Unlike RTDs (Resistance Temperature Detectors), which are made from metals, NTC thermistors are generally made of ceramics or polymers. Different materials used result in different temperature responses, as well as other characteristics.

Temperature response

While most NTC thermistors are typically suitable for use within a temperature range between -55°C and 200°C, where they give their most precise readings, there are special families of NTC thermistors that can be used at temperatures approaching absolute zero (-273.15°C) as well as those specifically designed for use above 150°C

As can be seen from the figure, the NTC thermistors have a much steeper resistance-temperature slope compared to platinum alloy RTDs, which translates to better temperature sensitivity.



7. What are thermal sensors? Explain any three thermal sensors. With neat circuit show how temperature is measured using thermocouple. List the merits and demerits of using Thermocouple. [9 marks] (December 2020)

Temperature detection is the foundation for all advanced forms of temperature control and compensation.

The temperature detection circuit itself monitors ambient temperature.

When a specific high temperature threshold is exceeded, preventative action can be taken by the system to lower the temperature. An example of this is turning on a fan.

Similarly, a temperature detection circuit can serve as the core of a temperature compensation function.

There are four commonly used temperature sensor types: RTD, NTC Thermistor, Thermocouples, Semiconductor based temperature sensors.

Resistance Temperature Detectors (RTDs)

Resistance temperature detectors, or resistance thermometers, employ a sensitive element of extremely pure platinum, copper or nickel wire that provides a definite resistance value at each temperature within its range.

The relationship between temperature and resistance of conductors in the temperature range near 0°C can be calculated from the equation

$$R_t = R_{ref} (1 + \alpha \Delta t)$$

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R_{ref} = resistance at the reference temperature, usually 0°C

α = temperature coefficient of resistance

Δt = difference between operating and reference temperature

Almost all metallic conductors have a positive temperature coefficient of resistance so that their resistance increases with an increase in temperature.

Some materials, such as carbon and germanium, have a negative temperature coefficient of resistance that signifies that the resistance decreases with an increase in temperature

A high value of α is desirable in a temperature sensing element so that a substantial change in resistance occurs for a relatively small change in temperature.

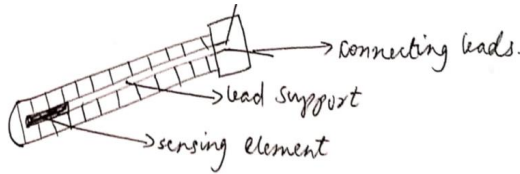
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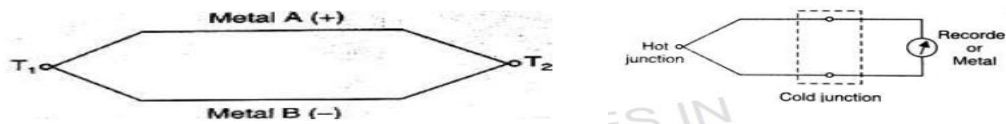
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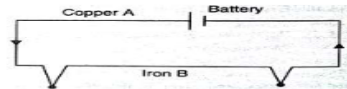
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Thompson effect.

- When a current flows through a conductor having thermal gradient along length of the conductor, heat is released at a point where current is in the direction same as the heat flow while heat is absorbed at a point where current flows in the direction opposite to that of heat flow.

Advantages and disadvantages of Thermocouple

Advantages:

- Rugged construction
- Cheap (cost)
- Speed of response high
- Measurement – accurate and bridge circuit not required

Disadvantages

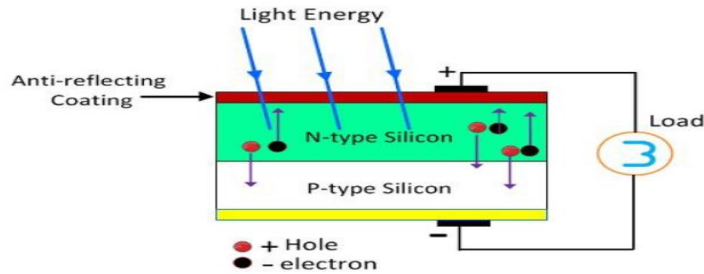
- Non-linearity in EMF vs temp. characteristics
- Signal needs to be amplified in many applications
- Cold junction and other compensation is essential for accurate measurement.

8. What is photovoltaic cell? Explain its construction and characteristics with neat figure.

[5 marks] (December 2020)

Photo-voltaic cell

The photovoltaic cell is the type of active transducer. The current starts flowing into the photovoltaic cell when the load is connected to it. The silicon and selenium are used as a semiconductor material. When the semiconductor material absorbs heat, the free electrons of the material starts moving. This phenomenon is known as the photovoltaic effect.

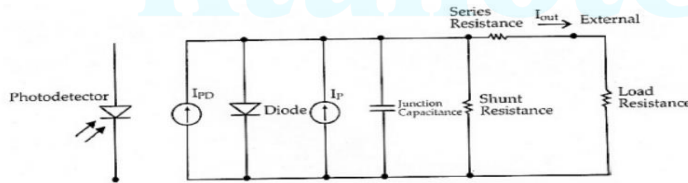


Photovoltaic Cell

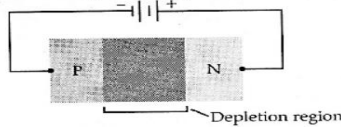
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